

PREFACE FOR FACULTY

IIR digital filter design typically transforms well-established continuous-time analog filter prototypes (Butterworth, Chebyshev) into discrete transfer functions. This lecture covers continuous analog filter theory, pole placement, and order calculation equations.

Pedagogical Strategy:

1. Formulate the **Butterworth Approximation**: Maximally flat magnitude response in passband and stopband:

$$|H_a(j\Omega)|^2 = \frac{1}{1 + (\Omega/\Omega_c)^{2N}}$$

2. Derive the Butterworth order formula N and 3-dB cutoff Ω_c .
3. Show pole placement on a circle of radius Ω_c in the left-half s -plane.
4. Formulate **Chebyshev Type I (Equiripple Passband)** and **Type II (Inverse Chebyshev)** approximations using Chebyshev polynomials $C_N(x)$.
5. Contrast Butterworth and Chebyshev prototypes: Chebyshev achieves a much steeper transition roll-off for the same filter order N .

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Calculate** required filter order N and cutoff Ω_c for Butterworth and Chebyshev analog lowpass prototypes.
2. **Determine** stable left-half s -plane pole locations for Butterworth and Chebyshev filters.
3. **Synthesize** continuous-time transfer functions $H_a(s)$ from given attenuation specifications.
4. **Compare** Butterworth, Chebyshev Type I, and Chebyshev Type II responses.

2. MATHEMATICAL FOUNDATIONS

2.1 Butterworth Filter Approximation

- **Magnitude Squared Function:**

$$|H_a(j\Omega)|^2 = \frac{1}{1 + \epsilon^2(\Omega/\Omega_p)^{2N}} = \frac{1}{1 + (\Omega/\Omega_c)^{2N}}$$

- **Order Formula:**

$$N \geq \frac{\log_{10} \sqrt{\frac{10^{0.1A_s}-1}{10^{0.1A_p}-1}}}{\log_{10}(\Omega_s/\Omega_p)} = \frac{\log_{10}(\lambda/\epsilon)}{\log_{10}(\Omega_s/\Omega_p)}$$

- **3-dB Cutoff Frequency Ω_c :**

$$\Omega_c = \frac{\Omega_p}{(10^{0.1A_p} - 1)^{1/(2N)}}$$

- **Pole Locations in s -Plane:** $2N$ poles uniformly spaced on circle of radius Ω_c . The N stable left-half plane poles are:

$$s_k = \Omega_c e^{j\frac{\pi}{2}[1+(2k-1)/N]}, \quad k = 1, 2, \dots, N$$

2.2 Chebyshev Type I Filter Approximation

- **Magnitude Squared Function:**

$$|H_a(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 C_N^2(\Omega/\Omega_p)}, \quad \epsilon = \sqrt{10^{0.1A_p} - 1}$$

- **Chebyshev Polynomials:**

$$C_N(x) = \begin{cases} \cos(N \arccos x), & |x| \leq 1 \\ \cosh(N \operatorname{arccosh} x), & |x| > 1 \end{cases}$$

Recursion: $C_{N+1}(x) = 2xC_N(x) - C_{N-1}(x)$ with $C_0(x) = 1, C_1(x) = x$.

- **Order Formula:**

$$N \geq \frac{\operatorname{arccosh}(\lambda/\epsilon)}{\operatorname{arccosh}(\Omega_s/\Omega_p)} = \frac{\ln \left[\frac{\lambda}{\epsilon} + \sqrt{\left(\frac{\lambda}{\epsilon}\right)^2 - 1} \right]}{\ln \left[\frac{\Omega_s}{\Omega_p} + \sqrt{\left(\frac{\Omega_s}{\Omega_p}\right)^2 - 1} \right]}$$

3. WORKED NUMERICAL EXAMPLES

Example 26.1: Analog Butterworth Lowpass Filter Design

Problem: Design an analog Butterworth lowpass filter to satisfy:

- Passband loss ≤ 1 dB at $\Omega_p = 100$ rad/s ($A_p = 1$ dB)
- Stopband attenuation ≥ 20 dB at $\Omega_s = 200$ rad/s ($A_s = 20$ dB)

Solution:

1. Calculate Order N :

- $\epsilon = \sqrt{10^{0.1(1)} - 1} = \sqrt{1.2589 - 1} = \sqrt{0.2589} = 0.5088$
- $\lambda = \sqrt{10^{0.1(20)} - 1} = \sqrt{100 - 1} = \sqrt{99} = 9.9499$
- Ratio $\frac{\lambda}{\epsilon} = \frac{9.9499}{0.5088} = 19.555$
- Frequency ratio $\frac{\Omega_s}{\Omega_p} = \frac{200}{100} = 2.0$

$$N \geq \frac{\log_{10}(19.555)}{\log_{10}(2.0)} = \frac{1.2913}{0.3010} = 4.29 \implies \mathbf{N = 5}$$

2. Calculate Cutoff Frequency Ω_c :

$$\Omega_c = \frac{\Omega_p}{\epsilon^{1/N}} = \frac{100}{(0.5088)^{1/5}} = \frac{100}{0.8735} = 114.48 \text{ rad/s}$$

3. Poles ($N = 5$):

- $s_1, s_5 = 114.48e^{j(180^\circ \pm 36^\circ)} = 114.48(-0.8090 \pm j0.5878) = -92.62 \pm j67.29$
- $s_2, s_4 = 114.48e^{j(180^\circ \pm 72^\circ)} = 114.48(-0.3090 \pm j0.9511) = -35.37 \pm j108.88$
- $s_3 = 114.48e^{j180^\circ} = -114.48$

4. Transfer Function $H_a(s)$:

$$H_a(s) = \frac{\Omega_c^5}{(s + 114.48)(s^2 + 185.24s + 13106)(s^2 + 70.74s + 13106)}$$

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Derive the expressions for the order N and cutoff frequency Ω_c of an analog Butterworth lowpass filter. (8 Marks) **(b)** Design an analog Chebyshev Type I lowpass filter with 1 dB passband ripple up to $\Omega_p = 10$ rad/s and stopband attenuation ≥ 30 dB at $\Omega_s = 25$ rad/s. Find order N . (7 Marks)

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):** Complete derivation from magnitude equations at Ω_p and Ω_s (8 Marks)
- **Part (b):**
 - $\epsilon = \sqrt{10^{0.1} - 1} = 0.5088, ; \lambda = \sqrt{10^3 - 1} = 31.607$
 - $\frac{\lambda}{\epsilon} = \frac{31.607}{0.5088} = 62.12$
 - $\frac{\Omega_s}{\Omega_p} = \frac{25}{10} = 2.5$
 - $N \geq \frac{\text{arcosh}(62.12)}{\text{arcosh}(2.5)} = \frac{\ln(62.12 + \sqrt{62.12^2 - 1})}{\ln(2.5 + \sqrt{2.5^2 - 1})} = \frac{4.822}{1.5668} = 3.078 \implies \mathbf{N = 4}$ (7 Marks)

5. PYTHON VERIFICATION SCRIPT

```
import scipy.signal as signal

N, Wn = signal.buttord(100, 200, 1, 20, analog=True)
print(f"Butterworth Order: {N}, Cutoff: {Wn:.2f} rad/s")
```